



SMART AUTOMATION VS PHYSICAL AI

● **Smart automation systems are designed to act based on pre-programmed data. There is no independent decision-making by the system, no built-in intelligence. Any deviation from the programmed algorithm requires human intervention. These work well for set tasks that involve a high degree of repetition. Also known as Industry 4.0.**

Physical AI adds an intelligence layer on top of smart automation that gives it cognitive capabilities to analyse the data and take an independent decision based on the inputs and not a pre-programmed algorithm. This can reduce human-machine interface to a significant extent

on jobs that are repetitive or unsafe for humans. But the intelligence layer is reducing the human-machine interface by giving the machine cognitive and perceptive capabilities to take decisions that earlier required human intervention. “Some jobs are hazardous or require extensive training. These companies are becoming leaders in AI adoption,” says Vijaya Ghosh, Partner, BCG. When it comes to areas such as predictive maintenance, the system is trained to think like a human operator and move away from a rule-based system, which is what is happening at companies like Tata Steel. Here, the system can read cues like abnormally high temperature or unusual sounds, analyse, and decide to shut down the system for safety reasons.



WHY WE NEED PHYSICAL AI

● **Persistent labour shortages, escalating cost pressures and fragile global supply chains, amplified by geopolitical and market uncertainty, are converging to threaten productivity, profitability and resilience. At the same time, growing consumer expectations for speed, customisation and sustainability demand operational flexibility.**

Advances in AI and complex simulations, enabled by accelerated computing, have made it feasible to run AI models and algorithms in real time. This enables robots to perceive, plan and act in complex, real-world scenarios, effectively achieving a level of physical intelligence

The marriage between AI and automation has created a digital super operator on the shop floor. Most factories have an old-time employee with a wealth of non-transferrable knowledge about the system who knows exactly how to handle each machine. Over time, training automation systems to act this intelligently will reduce the risk of dependence on one or two people capable of effectively managing the shopfloor. “We are asking clients to use AI to pick up that knowledge and become a smart operator who’s able to do all this and take cues not only from the experience of that person but also use data to come up with insights which didn’t exist before,” says Ghosh. This is one reason why AI-driven interfaces can be worth the investment.